2026 IEEE 8th International Conference on Communications, Information System and Computer Engineering (CISCE)

Unequal Error Protection Scheme Based on Implicit Partial Product-LDPC Codes

Xuemei Zhu[1], Jiawei Duan[2], Zhicheng Huang[1], Xiaodong Zhang[3], Kuan Wu[4], Qianfan Wang[4,*]
[1]*Academic Affairs Office, Guangdong Women's Polytechnic*, Guangzhou, China
[2]*School of Computer Science and Engineering, Sun Yat-sen University*, Guangzhou, China
[3]*Network Management Center, China Mobile Group Guangdong Company Limited*, Guangzhou, China
[4]*Department of Computer Science, City University of Hong Kong*, Hong Kong SAR, China
Email: 0340212@gdfs.edu.cn, jiawei.duan@connect.polyu.hk, zchuang@gdfs.edu.cn,
zhangxiaodong2@gd.chinamobile.com, {kuanwu, *qwang742}@cityu.edu.hk

2026 IEEE 8th International Conference on Communications, Information System and Computer Engineering (CISCE) | 979-8-3315-6042-3/26/$31.00 ©2026 IEEE | DOI: 10.1109/CISCE69494.2026.11504662

***Abstract*—This paper proposes an unequal error protection (UEP) scheme based on implicit partial product-LDPC codes. All rows of the source data array are encoded by a low-density parity-check (LDPC) code, while only a prescribed subset of information columns is further protected by a Bose–Chaudhuri–Hocquenghem (BCH) code. The check information generated by the column code is not transmitted explicitly. Instead, it is superimposed on the row LDPC codewords through a free-ride coding mechanism. In this way, the selected columns, referred to as the most important data (MID), are protected by both the row code and the column code, whereas the remaining columns, referred to as the less important data (LID), are protected only by the row code. A turbo-like iterative decoding algorithm is then adopted for the proposed scheme, consisting of free-ride decoding, row decoding, column decoding, and row re-decoding. We further present an approximate performance characterization for the doubly protected MID and a genie-aided lower bound for the singly protected LID. Simulation results show that the proposed scheme provides a substantial reliability gain for the MID, while the LID performance is not sacrificed and can even outperform that of standalone LDPC coding in the medium-to-high signal-to-noise ratio (SNR) region.**

***Index Terms*—Unequal error protection, implicit partial product-LDPC codes, free-ride coding.**

I. Introduction

In many practical communication systems, different portions of the transmitted data usually have different reliability requirements. A typical example is that the header of a packet is more important than the payload, since errors in the header may cause the whole packet to be discarded, whereas a few errors in the payload may still be tolerable at the application layer [1]–[3]. This naturally leads to the problem of unequal error protection (UEP), where the most important data (MID) should be protected more strongly than the less important data (LID).

Existing UEP techniques can be implemented either at the modulation level or at the coding level. Representative modulation-based approaches include multi-resolution or hierarchical modulation [4]–[6], multilevel modulation with unequal protection [7], and multiplexed hierarchical constellations [8]. On the coding side, early UEP constructions were studied in [9], [10]. Later developments include rate-compatible punctured convolutional codes and turbo codes [2], [11], UEP-LDPC codes based on degree-distribution design [12]–[14], Plotkin-type superposition constructions [15], and layered superposition methods based on block Markov superposition transmission or partial superposition transmission [16]–[19].

Although these approaches are effective, two difficulties often remain. First, improving the reliability of the MID usually requires additional redundancy or extra transmission resources. Second, the performance of the LID may be degraded when stronger protection is concentrated on the MID. These observations motivate us to revisit free-ride coding [20], which admits reliable transmission of a small number of extra bits over an existing LDPC-coded link without additional bandwidth expansion. Free-ride coding has been applied to code constructions and communication scenarios such as coupled LDPC codes [21]–[26], feedback information transmission [27], [28], and critical communication scenarios in vehicular networks [29], [30].

Closely related implicit product-code structures were originally introduced to improve the overall performance of LDPC-coded systems by embedding partial column checks into row codewords [21], [22]. In this paper, we study this coding structure from the perspective of UEP. The basic idea is to encode all rows by an LDPC code while imposing additional column-wise protection only on a prescribed subset of information columns. The column check information is then transmitted implicitly through free-ride coding. As a result, the selected columns receive double protection and the remaining columns receive single protection. More importantly, once a subset of the protected columns is successfully recovered, their reliable decisions can be fed back to improve the subsequent row decoding, so that the LID is not necessarily sacrificed. The main contributions of this paper are summarized as follows.

- We present a UEP scheme via implicit partial product-

This work was supported by the NSFC (No. 62301617), the Guangdong Basic and Applied Basic Research Foundation (No. 2025A1515011650), the Young Talent Support Project of Guangzhou Association for Science and Technology (No. QT-2025-048), and the Guangdong Provincial Educational Science Planning Project (No. 2025GXJK0820). (**Corresponding author: Qianfan Wang)*

LDPC codes, together with a turbo-like iterative decoder. In the proposed scheme, all rows are encoded by an LDPC code, while only a prescribed subset of information columns is further protected by a BCH code. The resulting column check information is superimposed on the row LDPC codewords through free-ride coding, thereby creating a natural separation between the doubly protected most important data (MID) and the singly protected less important data (LID). The decoder iterates among free-ride decoding, row decoding, column decoding, and row re-decoding, where successfully decoded MID columns are hardened and reused to improve row decoding.
- We develop an analytical framework to characterize the UEP performance. For the doubly protected MID, we derive approximate expressions for the word error rate (WER) and bit error rate (BER) based on a binary symmetric channel (BSC) model after row decoding. For the singly protected LID, we derive a genie-aided lower bound by assuming partial knowledge of the successfully recovered MID positions.
- Simulation results show that the proposed scheme provides a significant reliability gain for the MID, while incurring essentially no loss for the LID; in the medium-to-high SNR region, the LID can even outperform standalone LDPC coding. The analytical results are also shown to agree well with the simulations in this regime.

II. Preliminaries

A. Conventional Product Codes and Design Motivation

Let $\mathbb{F}_2 = \{0,1\}$ be the binary field. Let $\mathcal{C}[n,k]$ be a row code of length n and dimension k, and let $\mathcal{C}_1[n_1,k_1]$ be a column code of length n_1 and dimension k_1. A codeword of a conventional product code can be represented by an $n_1 \times n$ coded array, where each row is a codeword of the row code and each column is a codeword of the column code.

Conventional product codes are attractive because their minimum distance is usually much larger than that of either component code. However, this performance improvement comes at the price of extra redundancy. Specifically, compared with the row code rate k/n, the code rate of the conventional product code becomes $\frac{kk_1}{nn_1}$, which is strictly smaller than k/n whenever $k_1 < n_1$. In addition, all information columns are protected in the same manner, which corresponds to equal error protection rather than unequal error protection.

These observations motivate a different product-like construction. Instead of protecting all columns, we protect only a prescribed subset of information columns. Furthermore, to avoid explicit transmission of the additional column check bits, we exploit free-ride coding so that the extra column check information can be conveyed implicitly through the existing row LDPC codewords.

B. Free-Ride Coding

Free-ride coding is a recently proposed technique that enables transmission of a small amount of extra data over an existing LDPC-coded link without additional transmission resources [20]. Let the payload data be denoted by $\boldsymbol{u} = \left(\boldsymbol{u}^{(0)}, \boldsymbol{u}^{(1)}, \ldots, \boldsymbol{u}^{(L-1)}\right)$, where each $\boldsymbol{u}^{(i)} \in \mathbb{F}_2^k$. Let the extra data be denoted by $\boldsymbol{f} = \left(\boldsymbol{f}^{(0)}, \boldsymbol{f}^{(1)}, \ldots, \boldsymbol{f}^{(L-1)}\right)$, where each $\boldsymbol{f}^{(i)} \in \mathbb{F}_2^{\ell}$. Let $\mathcal{C}_{\mathrm{p}}[n,k]$ be an LDPC code with parity-check matrix $\mathbf{H}$. Encoding the payload block by block yields $\boldsymbol{v} = \left(\boldsymbol{v}^{(0)}, \boldsymbol{v}^{(1)}, \ldots, \boldsymbol{v}^{(L-1)}\right)$, where each $\boldsymbol{v}^{(i)} \in \mathbb{F}_2^n$ satisfies $\boldsymbol{v}^{(i)}\mathbf{H}^{\mathsf{T}} = \mathbf{0}$.

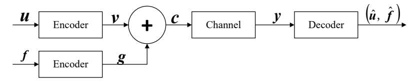

Fig. 1. Free-ride coding transmission model.

The extra data $\boldsymbol{f}$ is encoded by the free-ride encoder into a free-ride codeword $\boldsymbol{g} \in \mathbb{F}_2^{nL}$. The transmitted sequence is then formed as

$$\boldsymbol{c} = \boldsymbol{v} + \boldsymbol{g}, \tag{1}$$

where the addition is over $\mathbb{F}_2$. The free-ride coding transmission model is illustrated in Fig. 1. At the receiver, free-ride decoding is performed first to recover the extra data. After that, the influence of the estimated free-ride codeword can be removed from the log-likelihood ratio (LLR) sequence, and the payload LDPC code is then decoded.

Specifically, let $\boldsymbol{y}$ be the noisy observation of $\boldsymbol{c}$ and let $\boldsymbol{\Lambda}$ be the corresponding LLR sequence. After free-ride decoding, we obtain an estimate $\widehat{\boldsymbol{g}}$ and update the LLRs as

$$\widetilde{\Lambda}_i = (-1)^{\widehat{g}_i}\Lambda_i. \tag{2}$$

The updated LLR sequence $\widetilde{\boldsymbol{\Lambda}}$ is then used to decode the payload LDPC code.

III. Proposed UEP Scheme

A. Encoding and MID/LID Partition

Let the row code be an LDPC code $\mathcal{C}[n,k]$ with parity-check matrix $\mathbf{H}$, and let the column code be a high-rate algebraic code $\mathcal{C}_1[n_1,k_1]$ with parity-check matrix $\mathbf{H}_1$. Let the information array be denoted by $\boldsymbol{u} = \left(\boldsymbol{u}^{(0)}, \boldsymbol{u}^{(1)}, \ldots, \boldsymbol{u}^{(n_1-1)}\right)$, where each row vector $\boldsymbol{u}^{(i)} \in \mathbb{F}_2^k$. Hence, the information array has size $n_1 \times k$.

All rows are first encoded independently by the LDPC code, producing the row-code array $\boldsymbol{v} = \left(\boldsymbol{v}^{(0)}, \boldsymbol{v}^{(1)}, \ldots, \boldsymbol{v}^{(n_1-1)}\right)$, where each $\boldsymbol{v}^{(i)} \in \mathbb{F}_2^n$. Among the k information columns, we choose n_0 columns to receive additional protection. Without loss of generality, we take the first n_0 columns and denote them by $\boldsymbol{u}_0, \boldsymbol{u}_1, \ldots, \boldsymbol{u}_{n_0-1}$, where each $\boldsymbol{u}_j \in \mathbb{F}_2^{n_1}$. In this paper, the column check information is implemented as the column syndrome

$$\boldsymbol{s}_j = \boldsymbol{u}_j\mathbf{H}_1^{\mathsf{T}} \in \mathbb{F}_2^{n_1-k_1}, \qquad 0 \le j \le n_0 - 1. \tag{3}$$

These column syndromes are concatenated, with zero padding if necessary, into a sequence $\boldsymbol{f} \in \mathbb{F}_2^{\ell n_1}$, where ℓ denotes the number of extra bits that can be carried per row

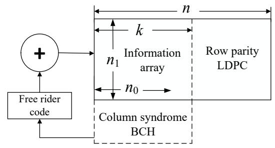

Fig. 2. Proposed UEP scheme based on implicit partial product-LDPC codes.

Algorithm 1 Encoding of the Proposed UEP Scheme

Input: Information array $\boldsymbol{u} \in \mathbb{F}_2^{n_1 \times k}$.
Output: Transmitted codeword $\boldsymbol{c} \in \mathbb{F}_2^{nn_1}$.

1: **for** $i = 0$ to $n_1 - 1$ **do**
2:     Encode row $\boldsymbol{u}^{(i)}$ by the LDPC code $\mathcal{C}[n, k]$ to obtain $\boldsymbol{v}^{(i)} \in \mathbb{F}_2^n$.
3: **end for**
4: **for** $j = 0$ to $n_0 - 1$ **do**
5:     Form the jth information column $\boldsymbol{u}_j \in \mathbb{F}_2^{n_1}$.
6:     Compute the column syndrome $\boldsymbol{s}_j = \boldsymbol{u}_j \mathbf{H}_1^{\mathsf{T}} \in \mathbb{F}_2^{n_1 - k_1}$.
7: **end for**
8: Concatenate $\boldsymbol{s}_0, \boldsymbol{s}_1, \ldots, \boldsymbol{s}_{n_0 - 1}$ and pad zeros if necessary to form $\boldsymbol{f} \in \mathbb{F}_2^{\ell n_1}$.
9: Encode $\boldsymbol{f}$ by the free-ride code to obtain $\boldsymbol{g} \in \mathbb{F}_2^{nn_1}$.
10: Form the transmitted codeword $\boldsymbol{c} = \boldsymbol{v} + \boldsymbol{g}$.

---

codeword by the free-ride link. Therefore, n_0 is chosen such that

$$n_0(n_1 - k_1) \leq \ell n_1. \tag{4}$$

The sequence $\boldsymbol{f}$ is encoded by the free-ride code into $\boldsymbol{g} \in \mathbb{F}_2^{nn_1}$, and the transmitted codeword is

$$\boldsymbol{c} = \boldsymbol{v} + \boldsymbol{g}. \tag{5}$$

By construction, the first n_0 information columns are protected by both the row LDPC code and the column BCH code. They are referred to as the *most important data* (MID). The remaining $k - n_0$ information columns are protected only by the row LDPC code and are referred to as the *less important data* (LID). This directly establishes the unequal-protection structure at the encoder.

The encoding procedure is summarized in Algorithm 1; see Fig. 2 for an illustration.

B. Turbo-Like Iterative Decoding

Let $\boldsymbol{y}$ be the noisy observation of $\boldsymbol{c}$ and let $\boldsymbol{\Lambda}$ be the corresponding LLR sequence. The decoding is organized in a turbo-like manner. We first decode the embedded column-check information through the free-ride decoder. After removing the estimated free-ride contribution, we decode the row LDPC code. The row decisions are then rearranged into columns, and the selected MID columns are decoded by the BCH decoder. Whenever a column is successfully decoded, the corresponding column positions in the row LLRs are hardened and fed back to the row decoder. This yields a second-round row decoding with improved reliability.

Algorithm 2 Turbo-Like Decoding of the Proposed UEP Scheme

Input: Received LLR sequence $\boldsymbol{\Lambda}$ associated with $\boldsymbol{y}$.
Output: Estimates of the MID and the LID.

1: Decode the free-ride code to obtain $\widehat{\boldsymbol{f}}$ and the corresponding estimate $\widehat{\boldsymbol{g}}$.
2: Update the LLRs according to (2) to obtain $\widetilde{\boldsymbol{\Lambda}}$.
3: Decode each row LDPC codeword using $\widetilde{\boldsymbol{\Lambda}}$ and obtain the first-round row decisions.
4: Rearrange the recovered information bits into columns.
5: **for** $j = 0$ to $n_0 - 1$ **do**
6:     Decode the jth MID column using the BCH decoder and the recovered syndrome $\widehat{\boldsymbol{s}}_j$.
7:     **if** the jth column is successfully decoded **then**
8:         Harden the corresponding column positions in the row LLRs.
9:     **end if**
10: **end for**
11: Perform a second-round row LDPC decoding using the updated LLRs.
12: Output the estimates of the MID and the LID.

The decoding schedule is summarized in Algorithm 2. Although the decoder can be iterated multiple times, the two-round schedule already captures the essential UEP behavior and is sufficient for the numerical results reported in this paper. The key role of the column decoder is not only to enhance the MID itself, but also to return reliable column decisions to the row decoder. This feedback is exactly the reason why the LID is not necessarily degraded and may even be improved.

IV. Performance Analysis

A. Approximate WER and BER of the MID

We begin with the MID, which is protected by both the row LDPC code and the column BCH code. Let $\mathrm{BER_p}$ denote the post-row-decoding BER of the original LDPC code. Let $\mathrm{BER_{pf}}$ denote the post-row-decoding BER in the presence of embedded column checks, and let $\mathrm{WER_f}$ denote the error rate of the free-ride decoder for the embedded column-check information. As suggested by the performance bounds of free-ride coding [27], in the medium-to-high SNR region we may use the approximation

$$\mathrm{BER_p} \leq \mathrm{BER_{pf}} \leq \mathrm{BER_p} + \mathrm{WER_f}. \tag{6}$$

When $\mathrm{WER_f} \ll \mathrm{BER_p}$, the input to the column decoder can be approximated by a BSC with crossover probability $\mathrm{BER_p}$.

Suppose that the BCH code can correct up to t errors. Then the column-wise WER of the MID can be approximated as [23] by

$$\mathrm{WER_{MID}} \approx \sum_{i=t+1}^{n_1} \binom{n_1}{i} \mathrm{BER}_{\mathrm{p}}^{i} \big(1 - \mathrm{BER_p}\big)^{n_1 - i}. \tag{7}$$

This approximation measures the probability that a protected MID column contains more than t bit errors after row decoding and therefore cannot be corrected by the BCH decoder.

We next estimate the BER of the MID. Following the same type of approximation used in related coupled-code analysis [31], when a protected column contains $i > t$ errors after row decoding, the resulting BER contribution can be upper-bounded by $\min\{(i+t)/k_1, 1\}$. Hence, the BER of the MID can be approximated as [23] by

$$\mathrm{BER}_{\mathrm{MID}} \approx \sum_{i=t+1}^{n_1} \binom{n_1}{i} \mathrm{BER}_{\mathrm{p}}^{i} (1-\mathrm{BER}_{\mathrm{p}})^{n_1-i} \min\left\{\frac{i+t}{k_1}, 1\right\}. \tag{8}$$

Equations (7) and (8) provide a simple analytical tool to predict the extremely low error rate of the MID in the region where direct Monte Carlo simulation becomes impractical.

B. Genie-Aided Lower Bound for the LID

We next consider the LID, which is protected only by the row LDPC code. When both the free-ride decoder and the column decoder are assumed to work perfectly, the best performance achievable for the LID is the row-decoding performance under partial knowledge of the correctly recovered MID positions. Based on this observation, we adopt a genie-aided (GA) approach [16], [17] to construct a lower bound for the BER of the LID.

The basic idea is to assume that approximately $n_0(1 - \mathrm{WER}_{\mathrm{MID}})$ MID columns are correctly recovered by the column decoder and are therefore known at the receiver. Let $\boldsymbol{v}_{\mathrm{L}}$ denote the corresponding known positions in the row codeword and let $\boldsymbol{v}_{\mathrm{R}}$ denote the remaining unknown positions. Partition the parity-check matrix of the row code as

$$\mathbf{H} = \left[\mathbf{H}_{\mathrm{L}}\ \mathbf{H}_{\mathrm{R}}\right]. \tag{9}$$

Then the remaining bits satisfy

$$\boldsymbol{v}_{\mathrm{R}} \mathbf{H}_{\mathrm{R}}^{\mathsf{T}} = \boldsymbol{v}_{\mathrm{L}} \mathbf{H}_{\mathrm{L}}^{\mathsf{T}}. \tag{10}$$

For a binary-input output-symmetric (BIOS) channel, (10) is equivalent to a standard syndrome-constrained BP decoding problem on a reduced Tanner graph. The corresponding BER, obtained by simulation under this known-bit assumption, serves as a genie-aided lower bound for the BER of the LID.

The bound is intentionally optimistic, but it is informative because it isolates the maximum possible benefit that the successfully decoded MID positions can bring to the LID. In the numerical results, this lower bound is shown to agree well with the practical performance of the LID in the medium-to-high SNR region.

V. Numerical Results

This section provides representative numerical results to illustrate the unequal-protection property of the proposed scheme.

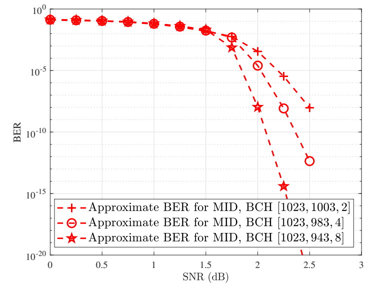

Fig. 3. Approximate BER of the doubly protected MID under different BCH codes.

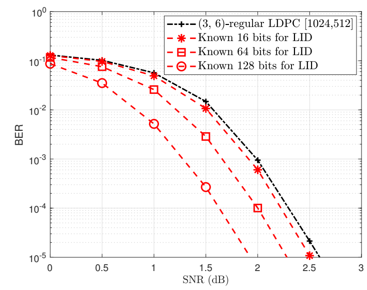

Fig. 4. Genie-aided BER lower bound for the singly protected LID.

A. MID Performance with Different BCH Codes

Example 1: Consider a rate-$1/2$, $(3, 6)$-regular LDPC code with parameters $[1024, 512]$, constructed by the progressive edge-growth (PEG) algorithm [32], as the row code. Figure 3 shows the approximate BER of the doubly protected MID under different BCH codes. As expected, a BCH code with stronger error-correcting capability yields a lower BER estimate for the MID. This confirms that the reliability of the MID can be flexibly adjusted by the choice of the column code.

B. Genie-Aided Lower Bound for the LID

Example 2: Consider the same $[1024, 512]$ row LDPC code as Example 1. Figure 4 shows the genie-aided BER lower bound for the LID under different numbers of known bits. We see that as the number of known bits increases, the BER lower bound decreases significantly, which is consistent with the intuition that more reliably recovered MID positions provide stronger assistance to the row decoder.

C. Comparison Between the MID and the LID

Example 3: Consider a rate-$1/2$, $(3, 6)$-regular LDPC code with parameters $[1024, 512]$ as the row code and a BCH code $[1023, 943]$ as the column code. In this case, the number of rows is $n_1 = 1023$. If each row LDPC codeword carries $\ell = 8$ extra bits, then the number of doubly protected columns is $n_0 = \left\lfloor \frac{1023\times 8}{80} \right\rfloor = 102$. Figure 5 compares the BERs of the original LDPC code, the doubly protected MID, and the singly protected LID. The approximate BER bound in (8) and the genie-aided lower bound for the LID are also shown. It can be observed that the BER of the MID is significantly lower

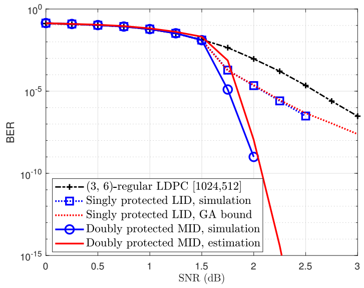

Fig. 5. BER comparison among the original LDPC code, the doubly protected MID, and the singly protected LID.

than that of the LID in the medium-to-high SNR region and closely follows the approximate bound. Moreover, the BER of the LID can outperform that of the original LDPC code and agrees well with the proposed genie-aided lower bound. In particular, around 2.25 dB, the approximate BER of the MID reaches the order of 10^{-15}, which indicates the extremely high reliability achievable for the protected information columns.

VI. Conclusion

This paper proposes an unequal error protection scheme based on implicit partial product-LDPC codes. All rows of the information array are encoded by an LDPC code, while only a subset of information columns is further protected by a BCH code. The resulting column-check information is conveyed implicitly through free-ride coding, so that the selected columns receive stronger protection without explicit transmission of additional column checks. We further adopted a turbo-like iterative decoder and present approximate expressions for the WER and BER of the MID, together with a genie-aided lower bound for the LID. Numerical results show that the proposed scheme provides a substantial reliability gain for the MID and can also improve the LID performance over standalone LDPC coding in the medium-to-high SNR region.

References

[1] C. Heegard, "Partitioned linear block codes for computer memory with "stuck-at" defects," *IEEE Trans. Inf. Theory*, vol. 29, no. 6, pp. 831–842, 1983.

[2] J. Hagenauer, "Rate-compatible punctured convolutional codes (RCPC codes) and their applications," *IEEE Trans. Commun.*, vol. 36, no. 4, pp. 389–400, 1988.

[3] *IEEE Std 802.11-2016, IEEE Standard for Information Technology—Telecommunications and Information Exchange Between Systems Local and Metropolitan Area Networks—Specific Requirements—Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications*, IEEE Std., 2016.

[4] L.-F. Wei, "Coded modulation with unequal error protection," *IEEE Trans. Commun.*, vol. 41, no. 10, pp. 1439–1449, 1993.

[5] K. Ramchandran, A. Ortega, K. M. Uz, and M. Vetterli, "Multiresolution broadcast for digital HDTV using joint source/channel coding," *IEEE J. Sel. Areas Commun.*, vol. 11, no. 1, pp. 6–23, 1993.

[6] M. Morimoto, M. Okada, and S. Komaki, "A hierarchical image transmission system for multimedia mobile communication," in *Proc. 1st Int. Workshop Wireless Image/Video Commun.*, 1996, pp. 80–84.

[7] R. H. Morelos-Zaragoza, M. P. C. Fossorier, S. Lin, and H. Imai, "Multilevel coded modulation for unequal error protection and multistage decoding—Part I: Symmetric constellations," *IEEE Trans. Commun.*, vol. 48, no. 2, pp. 204–213, 2000.

[8] S.-H. Chang, M. Rim, P. C. Cosman, and L. B. Milstein, "Optimized unequal error protection using multiplexed hierarchical modulation," *IEEE Trans. Inf. Theory*, vol. 58, no. 9, pp. 5816–5840, 2012.

[9] B. Masnick and J. Wolf, "On linear unequal error protection codes," *IEEE Trans. Inf. Theory*, vol. 13, no. 4, pp. 600–607, 1967.

[10] M.-C. Lin, C.-C. Lin, and S. Lin, "Computer search for binary cyclic UEP codes of odd length up to 65," *IEEE Trans. Inf. Theory*, vol. 36, no. 4, pp. 924–935, 1990.

[11] A. S. Barbulescu and S. S. Pietrobon, "Rate compatible turbo codes," *Electron. Lett.*, vol. 31, no. 7, pp. 535–536, 1995.

[12] N. Rahnavard and F. Fekri, "Unequal error protection using low-density parity-check codes," in *Proc. IEEE Int. Symp. Inf. Theory (ISIT)*, 2004, p. 449.

[13] N. Rahnavard, H. Pishro-Nik, and F. Fekri, "Unequal error protection using partially regular LDPC codes," *IEEE Trans. Commun.*, vol. 55, no. 3, pp. 387–391, 2007.

[14] X. Chen and F. C. M. Lau, "Optimisation of low-density parity-check codes with deterministic unequal error protection properties," *IET Commun.*, vol. 5, no. 11, pp. 1560–1565, 2011.

[15] V. Kumar and O. Milenkovic, "On unequal error protection LDPC codes based on plotkin-type constructions," *IEEE Trans. Commun.*, vol. 54, no. 6, pp. 994–1005, 2006.

[16] X. Ma, C. Liang, K. Huang, and Q. Zhuang, "Block Markov superposition transmission: Construction of big convolutional codes from short codes," *IEEE Trans. Inf. Theory*, vol. 61, no. 6, pp. 3150–3163, 2015.

[17] C. Liang, X. Ma, Q. Zhuang, and B. Bai, "Spatial coupling of generator matrices: A general approach to design good codes at a target BER," *IEEE Trans. Commun.*, vol. 62, no. 12, pp. 4211–4219, 2014.

[18] Q. Wang, S. Cai, W. Lin, S. Zhao, L. Chen, and X. Ma, "Spatially coupled LDPC codes via partial superposition and their application to HARQ," *IEEE Trans. Veh. Technol.*, vol. 70, no. 4, pp. 3493–3504, 2021.

[19] Q. Wang, S. Cai, W. Lin, L. Chen, and X. Ma, "Spatially coupled LDPC codes via partial superposition," in *Proc. IEEE Int. Symp. Inf. Theory*, 2019, pp. 2614–2618.

[20] S. Cai, S. Zhao, and X. Ma, "Free ride on LDPC coded transmission," *IEEE Trans. Inf. Theory*, vol. 68, no. 1, pp. 80–92, 2022.

[21] Q. Wang, S. Cai, and X. Ma, "Free-ride coding for constructions of coupled LDPC codes," *IEEE Trans. Commun.*, vol. 71, no. 3, pp. 1259–1270, 2023.

[22] Y. Wang, Q. Wang, and X. Ma, "Design of implicit partial product-LDPC codes and low complexity decoding algorithm," *IEEE Commun. Lett.*, vol. 27, no. 2, pp. 419–423, 2023.

[23] X. Ma, Q. Wang, S. Cai, and X. Xie, "Implicit partial product-LDPC codes using free-ride coding," in *Proc. IEEE Int. Conf. Commun.*, 2022, pp. 2471–2476.

[24] X. Ma, Q. Wang, M. Xie, and S. Cai, "Implicit globally-coupled LDPC codes using free-ride coding," in *Proc. IEEE Wireless Commun. Netw. Conf.*, 2022, pp. 1117–1122.

[25] Q. Wang, S. Cai, F. Meng, L. Chen, and X. Ma, "Partial product-LDPC codes without rate loss," in *Proc. IEEE/CIC Int. Conf. Commun. China (ICCC)*, 2022, pp. 208–213.

[26] F. Meng, Q. Wang, S. Cai, and X. Ma, "Parallel decoding of PIC-LDPC codes aided by free-ride coding," in *Proc. IEEE/CIC Int. Conf. Commun. China*, 2022, pp. 1003–1007.

[27] Q. Wang, S. Cai, Y. Wang, and X. Ma, "Free-ride feedback and superposition retransmission over LDPC coded links," *IEEE Trans. Commun.*, vol. 71, no. 1, pp. 13–25, 2023.

[28] Y. Wang, Q. Wang, M. Jiang, and X. Ma, "Improving performance of LDPC-coded links via partial superposition retransmission," *J. Commun. Inf. Netw.*, vol. 7, no. 3, pp. 259–268, 2022.

[29] M. Xie, J. Gong, Q. Wang, S. Cai, and X. Ma, "Reducing age of extra data by free riding on coded transmission in multiaccess networks," in *Proc. IEEE Wireless Commun. Netw. Conf.*, 2022, pp. 1296–1300.

[30] M. Xie, J. Gong, Q. Wang, and X. Ma, "Achieving two-level age by free-ride coding in preemptive mission-critical networks," in *Proc. IEEE Int. Conf. Commun. Workshops*, 2022, pp. 210–215.

[31] Y. Wang and X. Ma, "Rate-configurable SC-LDGM codes concatenated with zipper codes for optical communication," in *Proc. IEEE Annual Computing and Communication Workshop and Conference*, 2022, pp. 983–987.

[32] X.-Y. Hu, E. Eleftheriou, and D.-M. Arnold, "Regular and irregular progressive edge-growth Tanner graphs," *IEEE Trans. Inf. Theory*, vol. 51, no. 1, pp. 386–398, 2005.